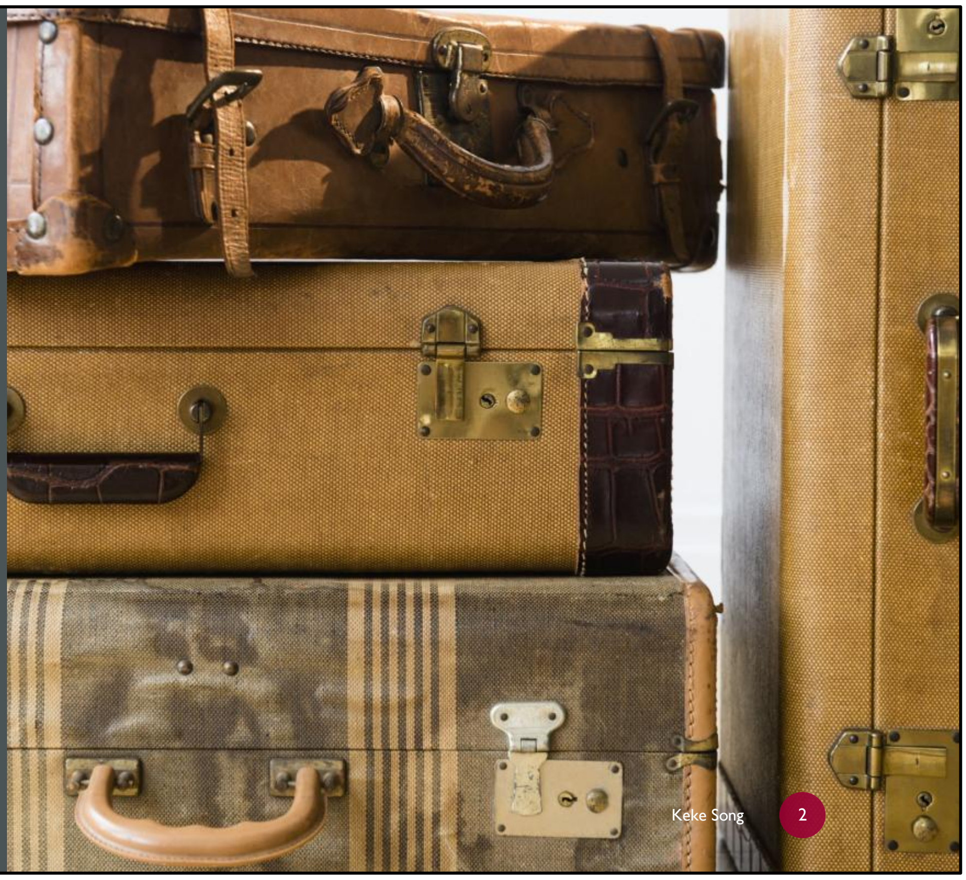
Module 4 :
Other issues in valuation

MINI CASES



Mini-Case A: Valuation of a privately held company
Issue #1: Estimate the cost of equity and WACC

- Private companies have no traded stocks to estimate the beta! What can we do?
  - We can use the betas of comparable (publicly traded) companies
  - commonly referred to as "pure play" approach

- In the mini case, we used company size (i.e., total assets or market cap), industry, product similarity, and geographic distribution to find the comparables.

- Discussion: What is the potential issue if we directly take the average estimated beta of these comparables?

1

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3

WACC
↓
RE
↓
β
↓
~~Return~~

A note on unlevered and levered betas:

$$\beta_{\text{levered}} = \beta_{unlevered} \times \left(1 + \frac{D}{E} \times (1 - T_c)\right)$$

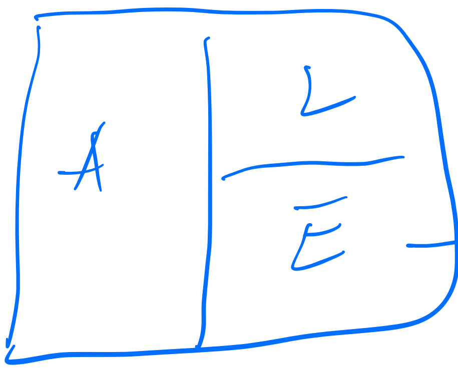
Unlevered beta depends on a company's
- business risk:
  - company size
  - assets composition
  - cost structure
  - products
  - geographic distribution
  - …….

Levered beta depends on a company's
- business risk:
  - company size
  - assets composition
  - cost structure
  - products
  - geographic distribution
  - …….
- financial leverage

- Based on the screening method of the mini case, the comparable companies should have a similar ________ beta.
- (a) unlevered; (b) levered

No financial leverage

$A \equiv E$



$\beta_2 = 1.2$

Sensity of stock return to mkt return

$\beta_u = \beta_A$

sensity of Asset return to mkt return

Valuation of Privately Held companies

Issue #1: Estimate the cost of equity and WACC: **

- Step 1: Find listed comparable companies with similar business risk to the private company.
- Step 2: Run regressions to estimate the equity beta (levered beta) of the comparable companies.
- Step 3: Calculate the unlevered betas of the comparable companies and calculate the average unlevered beta.

$$\beta_{\text{unlevered}} = \frac{\beta_{Levered}}{(1 + \frac{D}{E} \times (1 - T_c))}$$

- Step 4: Calculate the "relevered" beta using the private company's capital structure.

$$\beta_{\text{levered}} = \beta_{unlevered} \times \left(1 + \frac{D}{E} \times (1 - T_c)\right)$$

- Step 5: Estimate the private company's cost of equity and WACC.

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** Use supplemental Excel spreadsheet "week 3 private company valuation blank"

$24.77

Comparable

Adjusted ?

Listed

Issue # 2: Private Company Discount
evidence from U.S.

20% - 30% → $24.8

- Koeplin, Sarin and Shapiro (2005) find that American acquirers applied a significant discount for private targets.
  - Their valuation measure is *Enterprise Value*.
  - Their findings:
    - U.S. private targets are acquired at an average 20-30% discount in comparison with U.S. public targets.
    - International private targets are acquired at an average 40% - 50% discount in comparison with U.S. public targets.

- In the Airthread case we used in the previous class, the estimated discount for the private company is "**in the range of 35%, though rules of thumb often put the range in the area of 20% to 30%**".

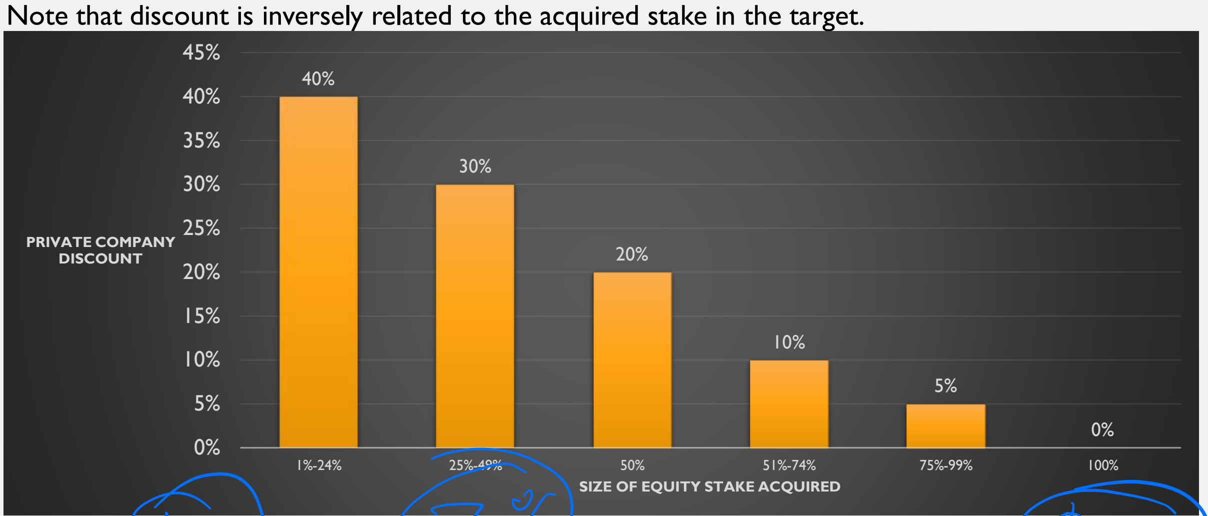
Private Company Discount
evidence from Australia

Acquirer $120

Offer price paid to private companies are at a discount compared to similar publicly *listed companies' market price*.

$100

Note that discount is inversely related to the acquired stake in the target.



$60 70% $100

Source: KPMG Valuation Survey 2015

'discount'

Q: What makes privately held companies different?

MF Superfund diversify β $R_E = r_f + \beta \times MRP$

Listed liquid ☺ transparency → track history

private less liquid ☹ less transparent → not much

founder's portfolio less diversities

Big α

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Valuation of multi-sector or conglomerate companies

- A conglomerate company is a large, often multinational, corporation that operates in different industry sectors.

- Question: in practice, it could be difficult to find comparable companies or industry peers for conglomerate or multi-sector targets. How can we deal with this issue?

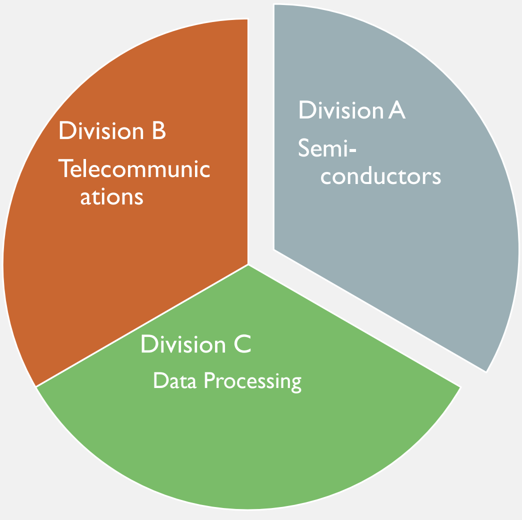
FCFF

Revenue forecast

$\frac{EV}{EBITDA}$

Comparable

Multiples Valuation
Conglomerate: "Sum-of-Parts Valuation"



- Conglomerate company is a "portfolio" of divisions operating in unrelated industries
- Different industries = different risk characteristics and growth prospects

- For each division, need to find its own set of comparable companies
  - value each division separately using its own multiple and sum up the divisional values together
- This approach is called "sum-of-parts valuation" (also referred to as "break-up value" approach)

Multiples Valuation
Conglomerate Valuation: An Example

- Diversity Ltd considers the company TechnoVision Inc. as a potential takeover target. TechnoVision has three divisions: semi-conductors, telecommunications, and data processing. Selected financials and other data for these divisions are in the table below.
- **Question:** given the data in the table, how much should Diversity Ltd bid for TechnoVision Inc?

Division	EBITDA	EV/EBITDA multiple (comparable companies)
Semi-conductors	$ 105.75 million	6.9
Telecommunications	$ 87.95 million	9.5
Data processing	$129.50 million	10.8

Multiples Valuation
Conglomerate Valuation: An Example (cont'd)

- **Answer:** using "sum-of-parts" valuation
  - *Estimated* EV of semi-conductors division: $6.9 \times \$105.75 \text{ million} = \$729.68 \text{ million}$
  - *Estimated* EV of telecommunications division: $9.5 \times \$87.95 \text{ million} = \$835.53 \text{ million}$
  - *Estimated* EV of data processing division: $10.8 \times \$129.50 \text{ million} = \$1398.60 \text{ million}$

- *Estimated* EV of TechnoVision Ltd.:

$$\$729.68 \text{ million} + \$835.53 \text{ million} + \$1398.60 \text{ million} = \$2963.81 \text{ million}$$

Conglomerate Discount

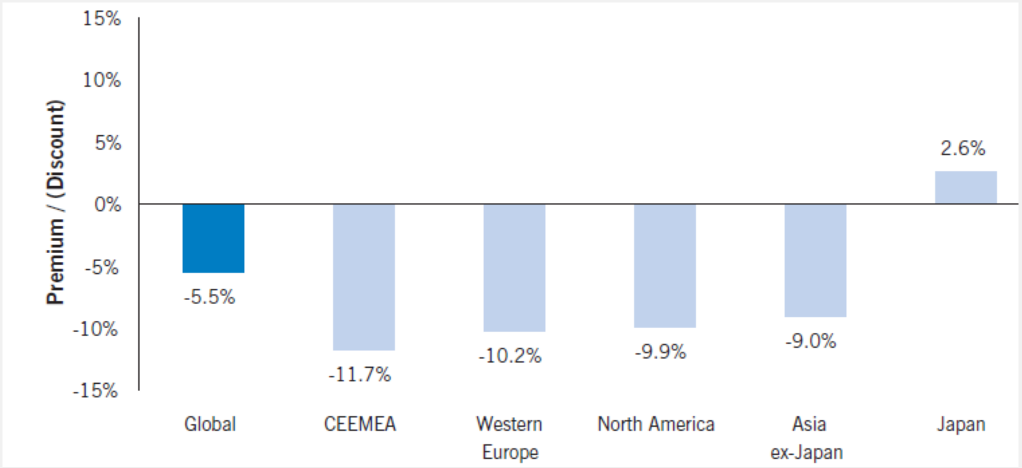
- From the previous example, **estimated** "sum-of-parts" value of TechnoVision Inc. enterprise is $2963.81 million.

- It is common that the **actual** enterprise value of TechnoVision is **below** the estimated "sum-of-parts" value.

- Question: Why are the market values of enterprise companies below their "fair values" (based on the "sum-of-parts")?

Conglomerate Discount

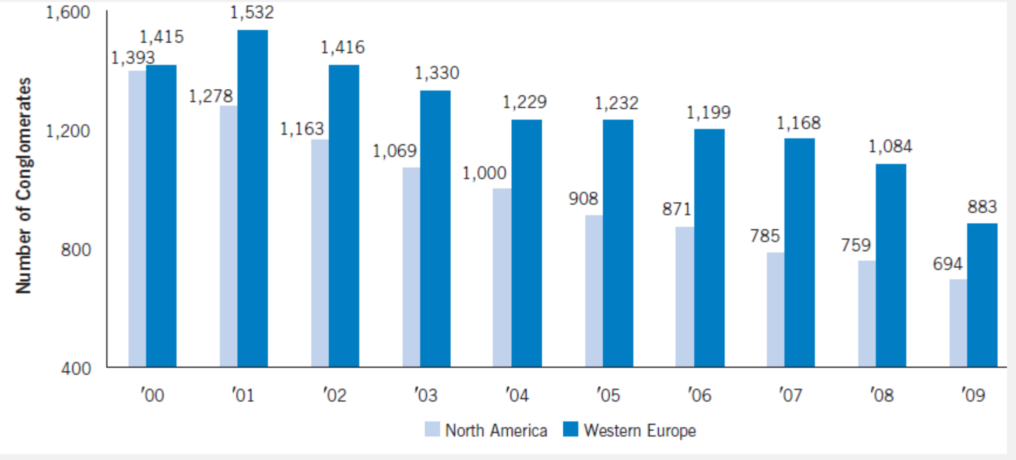
For the average conglomerate, it's market value is below it's "sum-of-parts" value. Investors appear to "penalize" conglomerate structure, creating a *conglomerate discount*.



Source: Shivdasani et al. (2011)

Conglomerate Discount, Cont'd

To increase value, conglomerates started to "break up" into separate companies.



Source: Shivdasani et al. (2011)

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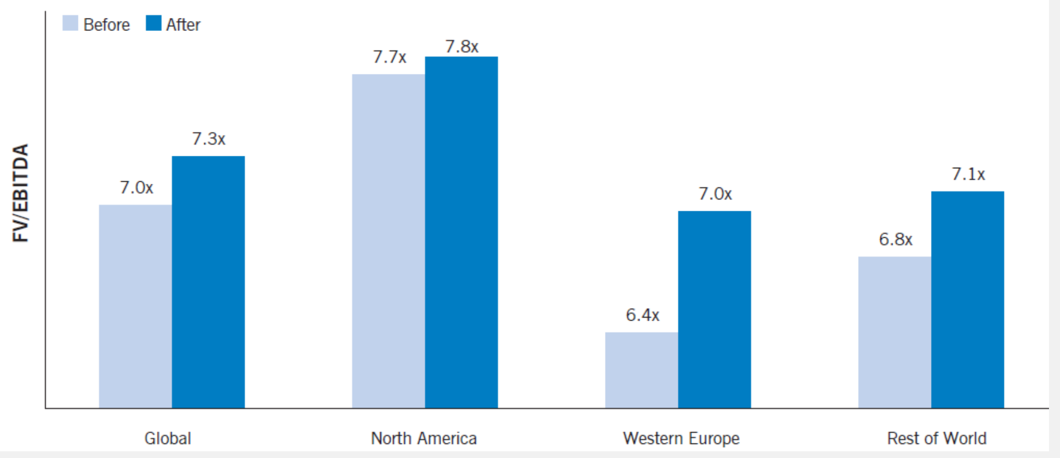
Breaking up a conglomerate seems to boost its value

Break-up of conglomerates into separate companies improved their multiples valuation



Source: Shivdasani et al. (2011)